

Ethan Maher

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Education

B.S., Computer Science

The University of Texas at Dallas

May 2025

Experience

Security Research Assistant

May 2025 - Current

The University of Texas at Dallas, Syssec Lab

- Developed and deployed a DNS data collection framework using C# to 80 machines to map IP addresses from logged system events to domain names, aiding system provenance graph analysis for machine learning models
- Partnered with a team of 5 student researchers to perform static analysis of Python malware samples to identify and understand payload behavior, improving collective accuracy of malware detection models

Projects

Todo-Sidebar.nvim | github.com/ethanamaher/todo-sidebar.nvim

May 2025

- Neovim plugin to recursively scan a Git repository for TODO, FIXME, and other customizable keywords, presenting results in a sidebar window
- Utilizes git grep for efficient file scanning inside files tracked by git

Better-Git-Blame.nvim | github.com/ethanamaher/better-git-blame.nvim

May 2025

- Developed a Neovim plugin for enhanced git history visualization, allowing users to analyze the commit history of a selected code block
- Leveraged git log's regex search to extract relevant diffs across different commits using Telescope and provide interactive diff views

Attendance Tracker | github.com/ethanamaher/AttendanceAppProject

Jan - May 2025

- Developed and designed a secure backend using C# and .NET for a Blazor full-stack class attendance tracking solution
- Engineered a MySQL database to manage and protect sensitive information for students and faculty, class schedules, and real-time attendance records

TUIpe Racer | github.com/ethanamaher/tuiperacer

Oct 2024

- Designed and built a terminal-based typing test in Go to track typing speeds
- Integrated a MySQL database to persistently store high scores, enabling leaderboard tracking and user progress analysis

Multi-Neighborhood Cellular Automata Simulator | github.com/ethanamaher/mnca_simulator

Sep 2024

- Developed a Golang simulation to model multi-neighborhood cellular automata using the Ebiten 2D game engine
- Optimized simulation performance through concurrent state calculations using goroutines

Volunteering and Leadership

Tex Mechs Keyboard Roundup

Aug 2023 - Apr 2024

Event Coordinator

- Co-organized a community technology convention for over 400 attendees, managing volunteer logistics and event flow at the University of Texas at Dallas
- Directed a team of 12 volunteers on the day of the event, delegating tasks for registration, setup, and attendee support

Skills

Programming Languages: C/C++, C#, Java, C, Python, Golang, Lua

Technologies: Bash, Git, Adobe Creative Cloud, Visual Studio, Linux, VMWare, Proxmox, Postgresql